

Learning from Examples

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Debug the Cause

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

If/Then Decisions

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Categories and Labels

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Explain a tied fit for Fitting a pattern from examples. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Fitting a pattern from examples. Proposed explanation: “The lowest error count is added across candidates”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Fit a new distance task

Imaginary map markers have distances 1, 3, 5 and 7 units, with supplied labels near, far, far and far. Compare candidates that predict far when distance is at least 2, 4 or 6 units. List every prediction and error count, select the lowest-error candidate, then predict at distance 2 before any comparison label is supplied. Limit your conclusion.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Training cards show lengths 2,3 labelled S and 7,8 labelled L. Two proposed rules are 'L at 5 or above' and 'L at 7 or above'. Both fit training. Choose one new labelled card length that would distinguish them and explain what remains unknown.

6. Solve the given case

Training examples label smooth leaves S and toothed leaves T. A new leaf is toothed.

Predict and state the evidence limit.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A learner sees three red apples and no green apples, then calls a green apple not an apple. What missing evidence could help? How would you check improvement?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 13 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Create the first small labeled example set for the sorter and explain which visual differences it covers.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Training cards show lengths 2,3 labelled S and 7,8 labelled L. Two proposed rules are 'L at 5 or above' and 'L at 7 or above'. Both fit training. Choose one new labelled card length that would distinguish them and explain what remains unknown.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Count each candidate from zero

Heights 2, 4, 6 and 8 cm have labels short, short, tall and tall. Predict all four labels under threshold 3, then under threshold 7, using tall when height is at least the threshold. List the disagreeing examples and separate totals.

My reasoning and evidence

13. W2 Explain a tied fit

Thresholds 5 and 6 both have zero disagreements on the W1 fitting set. The tie rule chooses the smaller threshold. Which is selected? Give a whole-number height where the two candidates disagree and both predictions. Explain what the tie rule does not prove.

My reasoning and evidence

14. W3 Update with an added example

Add height 5 cm labelled short to the four fitting examples. Recompute the disagreement totals for thresholds 3, 5, 6 and 7. Select the candidate with the fewest disagreements, keeping the smaller-threshold tie rule. Explain why the new label is part of fitting.

My reasoning and evidence

15. W4 Fit a new distance task

Imaginary map markers have distances 1, 3, 5 and 7 units, with supplied labels near, far, far and far. Compare candidates that predict far when distance is at least 2, 4 or 6 units. List every prediction and error count, select the lowest-error candidate, then predict at distance 2 before any comparison label is supplied. Limit your conclusion.

My reasoning and evidence

16. W5 Keep conflicting records visible

Two fitting records both have height 6 cm. One is labelled short and the other tall. A learner says a clever height-only rule can agree with both on every run. Explain the problem for a deterministic rule and name two checks to make before changing either record.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Learning from Examples

A fictional classifier learns whether measured sticks are short or long.

Training lengths 2,4,8,10 cm have labels short,short,long,long.

Candidate boundaries: long at least 5 or at least 7. Lowest errors wins; tie chooses smaller boundary.

Same training errors, different new prediction

	Boundary	Training errors	Prediction at 6
Candidate 5	5	0	long
Candidate 7	7	0	short

Read a cell using both its row and column labels.

Learning from Examples

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Training rows

2 and 4 are short; 8 and 10 are long.



2. Candidate checks

Boundaries 5 and 7 both predict S,S,L,L: zero errors each.



3. Tie policy

Choose boundary 5 because the declared tie rule prefers the smaller value.



4. New prediction

Length 6 is predicted long by the selected rule.

CHANGE ONE THING

Reverse the tie policy to prefer the larger boundary.

Explain the changed result using the matching lesson.

Which new measured length would reveal disagreement between the candidates?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Save candidate definitions, training errors and tie policy with any sorter model version.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Save candidate definitions, training errors and tie policy with any sorter model version.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional classifier learns whether measured sticks are short or long.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Length 6 is predicted long by the selected rule.

1. Training rows
2. New prediction
3. Tie policy
4. Candidate checks

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: New prediction Length 6 is predicted long by the selected rule. There is no training stick between 4 and 8 to distinguish these candidates. The tie policy chooses one consistently; it is not additional evidence that its prediction at 6 is true. Label-free prediction applies the selected boundary, while evaluation would require an appropriate new reference label.

Practice 2: explain the deciding evidence

Changed-case practice: Reverse the tie policy to prefer the larger boundary. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Boundary 7 is selected and predicts short at 6.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Boundary 7 is selected and predicts short at 6. The training fit is unchanged; only the explicit choice between tied candidates changes.

Repair a missing building block

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose input 0 and give both predicted outputs without inventing an observed result.. Return to this week and explain what you changed.

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which input detects the difference between the two written conditions? Explain its entire run.. Return to this week and explain what you changed.

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit evidence and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
